Infinite dessin orders, projective resolutions and spectral determinants

Extended proofs, Schur functors and Auslander–Reiten calculations

Research working draft

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Abstract

We work throughout with infinite-dimensional locally gentle algebras and completed surface orders arising from explicit cellular clean dessins on compact Riemann surfaces. We construct their projective resolutions, Cartan determinants and dual dessins; develop determinant semi-invariants, Schur modules and cyclic-quiver character kernels; and calculate actual lattice almost-split sequences and an affine-type diamond quotient. A constrained-path Fock algebra retains the gentle relations and has a compact extension with incidence boundary map. Its degree spectral triple and an inserted arithmetic Fredholm family are explicit. A pair of complex-isomorphic elliptic Belyi maps with different rational Frobenius factors isolates the necessary descent datum. Recent spectral approaches are compared with their actual hypotheses. Selected power-series, module and matrix statements are verified in Lean4. The arithmetic positive determinant comparison remains open; no proof of GRH is claimed.

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1 Cellular dessins and their infinite orders

We start with the surface-order constructions of Schreiber [1, 2], retaining the original sources unchanged. The generic-resolution source [3] is background, not a substitute for exactness of a particular complex. All primary algebra examples below come from explicit clean Belyi maps. Finite-dimensional objects occur as modules or specified quotient strata of those infinite algebras, not as replacements for them. The initial arithmetic target is a primitive nonprincipal Dirichlet L-function. Finite-order Hecke characters, Dedekind zeta functions and general Artin representations require separate comparison theorems; Artin holomorphy is not assumed.

A clean dessin on a compact oriented surface is specified by a nonempty finite dart set H, a fixed-point-free involution α, and a permutation σ such that $\langle \alpha, \sigma \rangle$ is transitive. Contract the white vertices of degree two. The resulting cellular graph has vertices the σ-orbits, edges the α-orbits, and faces the ϕ-orbits, where

$$\tau = \sigma^{-1}, \qquad \phi = \alpha\sigma^{-1}, \qquad 2 - 2g = \#\operatorname{cyc}(\sigma) - |H|/2 + \#\operatorname{cyc}(\phi).$$

Write $[h] = \{h, \alpha h\}$. Our quiver has vertices these edges and arrows $a_h : [h] \to [\tau h]$. Paths are composed in travel order: ab traverses a first. Set

$$A = kQ/(a_h a_{\phi h} : h \in H), \quad J = A_{>0}, \quad \Lambda = \varprojlim A/J^m.$$

Each vertex has two incoming and two outgoing arrows. Each arrow has precisely one permitted successor $a_{\tau h}$ and one forbidden successor $a_{\phi h}$; these are distinct because α has no fixed points. The same holds for predecessors. This is the *locally gentle* condition. Every permitted cycle repeats indefinitely, so A is infinite dimensional. We reserve "gentle" without qualification for the convention additionally imposing finite dimension.

Proposition 1.1 (The actual split order)**.** *For a permitted cycle c of length ℓ, let B_c be the hereditary cycle order in $M_\ell(k[[u_c]])$: diagonal and forward matrix entries are in $k[[u_c]]$, and entries crossing the chosen cycle cut are in $u_c k[[u_c]]$. Then Λ is the suborder of $B = \prod_c B_c$ whose two diagonal constant terms labelled $h, \alpha h$ agree. With $R = k[[u]]$ acting by $u \mapsto (u_c I)_c$, it is finite free of rank $\sum_c \ell(c)^2$; the conductor contains uB.*

Proof. In a cycle order a path of length less than ℓ is the corresponding matrix unit, multiplied by u_c exactly when it crosses the cut; another full turn multiplies it by u_c. Thus completed paths in each permitted branch give all entries of B_c. An arrow belongs to only that branch, whereas $e_{[h]}$ has diagonal entries at both occurrences $h, \alpha h$. Hence the only condition on branchwise formal coefficients is the displayed equality of diagonal constants. This proves both injectivity and surjectivity of the path-normal-form identification. Multiplication respects it: switching branches across a paired edge is precisely a forbidden pair. The quotient B/Λ is killed by u, so the submodule has the same generic rank as B. It is torsion free and finite over the DVR R, hence free of that rank. The path and u-adic topologies agree: $uA \subset J$, and if L is the longest permitted cycle, any path of length at least $L+1$ is u times a shorter nonempty path. Thus $J^{L+1} \subset uA$. This also justifies the completion used in the identification. □

The path-degree variable q below is not u: one full turn has path degree $\ell(c)$ but u-degree one. Formal, Laurent and operator-norm completions will likewise remain distinct.

Here are five explicit global inputs. Permutations list darts starting at zero; $E = |H|/2$.

Surface and Belyi map β	σ	α	$(g, \operatorname{rank}_R \Lambda)$
$\mathbb{P}^1$, $4z(1-z)$	$(0)(1)$	(01)	$(0,2)$
$\mathbb{P}^1$, $(z+1)^2/(4z)$	(01)	(01)	$(0,4)$
$y^2 = x^3 - 1$, x^3	$(012)(345)$	$(03)(14)(25)$	$(1,18)$
$\mathbb{P}^1$, $4z^3/(1+z^3)^2$	$(012)(345)$	$(03)(15)(24)$	$(0,18)$
$y^2 = x^3 - x$, x^2	(0123)	$(02)(13)$	$(1,16)$

The first is the nodal interval: $A = k[x,y]/(xy)$ and $\Lambda = k[[x,y]]/(xy)$. The second has $A = k\langle x,y\rangle/(x^2,y^2)$, still infinite dimensional. The two theta dessins have the same two permitted three-cycles but different face cycles, respectively (051324) and $(04)(13)(25)$. The elliptic square has one permitted and one forbidden four-cycle. For example x has pole order two on the cubic $y^2 = x^3 - 1$; x^3 has ramification $(3,3)$ above zero, $(2,2,2)$ above one since $x^3 - 1 = y^2$, and (6) above infinity. The rational theta map has passport $(3,3),(2,2,2),(2,2,2)$. These ramification computations, and the analogous direct zeros/poles in the other rows, verify cellular Belyi inputs rather than merely abstract quivers.

2 Projective resolutions, dual dessins and determinants

Use right modules $P_i = e_i A$, $S_i = P_i/P_i J$. Internal shift $M\langle n\rangle$ places its old generator in degree n. Let $\mathcal{R}_n(i)$ be the *indexing words* of length n from i with every adjacent pair forbidden, including e_i for $n = 0$. Such a word is usually zero in A; it still labels a summand.

Theorem 2.1 (Exact resolutions)**.** *There is a minimal graded projective resolution, with $P_0(i) = e_i A$ and for $n \geq 1$,*

$$P_n(i) = \bigoplus_{p = a_1 \cdots a_n \in \mathcal{R}_n(i)} e_{t(a_n)} A\langle n\rangle, \qquad d_n(x)_p = a_n x \text{ in the summand } a_1 \cdots a_{n-1}.$$

The augmentation $P_0(i) \to S_i$ is the quotient. For the completed order the identical finite-rank projective terms and formal-series maps resolve the residue simple. Ext and Tor against residue simples count forbidden words with the prescribed endpoints. In the graded case Ext has matching internal shift n.

Proof. The maps are right linear and degree zero. Adjacent differentials multiply by a forbidden pair. The augmentation kernel is the direct sum of aA over initial arrows. For any arrow a, multiplication $e_{t(a)}A \to e_{s(a)}A$ kills exactly the permitted paths beginning with its forbidden successor $f(a)$. All other basis paths survive to distinct basis paths. Thus its kernel is $f(a)A$. For a fixed prefix, different last arrows have disjoint surviving supports; there is no cancellation between summands. This proves exactness inductively. For formal series the same assertion holds coefficient by coefficient and identifies the entire closed kernel, not just its dense polynomial subspace. Terms are idempotent projectives and all entries are arrows. Consequently applying Hom or tensor with a residue simple kills every differential, proving minimality and the Ext/Tor count. In A minimality is graded; it does not require that J be its ordinary Jacobson radical. □

The two arms never terminate in this closed class. Their labelled patterns follow ϕ-cycles. An ungraded positive syzygy tail can have a shorter period by permutation of summands: the nodal interval has labelled period two but unordered tail period one; the elliptic theta has labelled period six and unordered tail period three. A simple need not be isomorphic to a positive syzygy; for the node it is R-torsion, whereas the positive syzygies are lattices.

Let $C_{ij}(q) = \sum_{d\geq 0} \dim_k(e_i A_d e_j)q^d$ and $E_{ij}(q) = \sum_{n\geq 0}(-1)^n \dim_k \operatorname{Ext}^n_A(S_i, S_j)q^n$. Degreewise Euler characteristic of the resolution gives $EC = I$: every coefficient is finite, because a summand in homological degree n starts in internal degree n. Constant terms are I, so the inverse is two-sided. Already signed $E(q)$ must not be evaluated at $-q$ again.

Proposition 2.2 (Dual dessin and cycle determinant)**.** *The complementary-relation algebra is another closed dessin algebra, with $\alpha^{\#} = \alpha$, $\sigma^{\#} = \phi^{-1}$, permitted successor ϕ, and forbidden successor τ. It exchanges graph vertices and faces and preserves genus. Its Cartan matrix satisfies $E(q) = C_{A^{\#}}(-q)$, with the opposite convention if required for Yoneda composition. For independent commuting arrow weights w_a,*

$$\det C_A(\mathbf{w}) = \frac{\prod_{f\in \mathrm{cyc}(\phi)}(1-(-1)^{|f|}w(f))}{\prod_{c\in \mathrm{cyc}(\sigma)}(1-w(c))}.$$

Proof. $\alpha(\sigma^{\#})^{-1} = \alpha\phi = \tau$, so the claimed relations and face data follow directly; the connectedness group is unchanged. To prove the determinant, put $S_{v,a} = w_a$ when $s(a) = v$, and $T_{a,v} = 1$ when $t(a) = v$, zero otherwise. On arrow states let $U_{a,b} = w_b$ for permitted adjacency and $V_{a,b} = w_b$ for forbidden adjacency. Then $TS = U + V$ and the path sum is $C = I + S(I - U)^{-1}T$. Sylvester's identity gives $\det C = \det(I + V)/\det(I - U)$. Each transition is a weighted permutation, whose cycle determinant is respectively $1 - (-1)^{\ell}w(c)$ and $1 - w(c)$. Everything holds in formal power series, hence as rational functions. This supplies a direct proof of the Bessenrodt–Holm formula in the closed dessin class [4]. □

The five univariate determinants are, in the order above,

$$\frac{1+q}{1-q}, \quad \frac{1+q}{1-q}, \quad \frac{1-q^6}{(1-q^3)^2}, \quad \frac{(1-q^2)^3}{(1-q^3)^2}, \quad 1.$$

The last cancellation does not imply finite global dimension: its forbidden resolution is infinite. These rational functions have cycle poles/zeros, not arithmetic prime Euler factors.

For clarity, a complex is a sequence with $d^2 = 0$, its homotopy category identifies maps differing by $dh + hd$, and $D(A)$ inverts maps inducing cohomology isomorphisms. A perfect complex is represented by a bounded complex of finite projectives. Our infinite simple resolutions compute derived Hom and tensor but are generally not perfect. The two-term presentation $[P_1 \to P_0]$

is perfect even if its kernel is nonzero. Quadratic linearity is a Koszul statement; it is not by itself an equivalence between all completed derived categories. Likewise a finite-dimensional gentle graded-curve theorem [6] cannot simply be transferred to these closed infinite orders. Here the surface walks explicitly compute the resolution endpoints: τ turns round a graph vertex and ϕ follows a face. For longer string objects all maps can be computed from the occurrence matrices; a universal geodesic-intersection classification for this completed category remains a separate theorem.

3 Determinant semi-invariants and Schur constructions

Assume k algebraically closed of characteristic zero for invariant-ring statements. A finite-dimensional representation of the *infinite* A has spaces N_i, arrow matrices $N(a)$ satisfying the monomial relations, and base-change group $G = \prod_i \mathrm{GL}(N_i)$. With column vectors, $N(a) \mapsto g_{t(a)} N(a) g_{s(a)}^{-1}$. Let $P_0 = \bigoplus P_i^{a_i}$, $P_1 = \bigoplus P_i^{b_i}$, and $f : P_1 \to P_0$ be a finite path-matrix map. When $\sum a_i \dim N_i = \sum b_i \dim N_i$, define

$$c_f(N) = \det D_f(N), \qquad D_f(N) = \mathrm{Hom}_A(f, N).$$

It is intrinsically a determinant-line section; chosen bases give this scalar.

Proposition 3.1 (Weights and exact homological scope)**.** $c_f(gN) = \prod_i \det(g_i)^{b_i - a_i} c_f(N)$. *Restricted determinants of this form span all semi-invariants on a closed G-stable relation or nilpotence locus. Set $M = \operatorname{coker} f$. On the square locus,*

$$c_f(N) \neq 0 \quad \Longleftrightarrow \quad \mathrm{Hom}(M, N) = 0 \quad \Longleftrightarrow \quad \mathrm{Hom}(M, N) = \mathrm{Ext}^1(M, N) = 0.$$

If f is injective, $\operatorname{coker} D_f = \mathrm{Ext}^1(M, N)$ *and higher Ext vanish. Without injectivity neither that cokernel equality nor higher vanishing follows.*

Proof. Naturality gives $D_f(gN) = G_1 D_f(N) G_0^{-1}$, where G_j repeats the relevant g_i. Take determinants. The contragredient coordinate-ring action has the inverse character. Schofield–van den Bergh's spanning theorem allows oriented cycles [36, Theorem2.3]. Exactness of invariants for reductive G in characteristic zero lifts a fixed-weight function from any closed stable locus to the ambient representation space; restrict the spanning determinants. Injectivity of the path map need not survive that restriction. Finally $\ker D_f = \mathrm{Hom}(M, N)$. Writing $K = \operatorname{im} f$, the exact sequence with projective P_0 expresses $\mathrm{Ext}^1(M, N)$ as $\mathrm{Hom}(K, N)/\operatorname{im} \mathrm{Hom}(P_0, N)$; the epimorphism $P_1 \to K$ embeds this quotient in $\operatorname{coker} D_f$. A square finite matrix with zero kernel has zero cokernel. If f is injective it is the actual length-one resolution, giving the stronger assertion. □

For the node, $f = \times x$, $M = \Lambda/(x)$, $N = \Lambda/(x, y^2)$ give $\dim \operatorname{coker} D_f = 2$ but $\dim \mathrm{Ext}^1(M, N) = 1$. For the elliptic square name arrows so that $a, b : 0 \to 1$, $c, d : 1 \to 0$, permitted cycle *acbd* and forbidden cycle *adbc*. With $f = a$ and $N(a) = 1$, $N(b) = N(c) = N(d) = 0$, its Hom matrix is [1], yet the subsequent forbidden resolution gives $\mathrm{Ext}^{2,3}(M, N) = k$. Thus nonvanishing is not all-degree derived orthogonality.

Proposition 3.2 (What completion does to invariants)**.** *A continuous d-dimensional representation of Λ kills J^d. On the reduced nilpotent representation variety of fixed dimension vector every invariant regular function is constant. Nonzero-weight semi-invariants need not be constant.*

Proof. Continuity with discrete finite-dimensional target kills some J^m; its Loewy filtration has at most d nonzero strict steps. Choose vertexwise complements to this filtration. An arrow sends a layer p into layers $q \geq p + 1$. Scaling layer p by z^p conjugates each component by z^{q-p}. At $z = 0$ the

representation tends to the zero-arrow representation. Every invariant therefore has the same value at all points; reducedness and the Nullstellensatz give the ring assertion. In the square example the scalar $N(a)$ varies on a nilpotent locus and has weight g_1/g_0, so semi-invariants survive. Formal entries in a projective map are evaluated on this locus by truncating modulo J^d, a legitimate finite polynomial operation. □

On the node, $u = x+y$ is regular and A is free of rank two over $k[u]$. The polynomial presentation $\times(u-\lambda)$ for $\lambda \neq 0$ is injective and its determinant on N is $\det(u_N - \lambda I)$, the characteristic polynomial of u_N up to the sign $(-1)^d$. But $u-\lambda$ is a unit in Λ, with its geometric-series inverse, so those modules do not extend continuously to the completion. An R-free lattice and its u-nilpotent quotients are different categories. On a lattice a separately specified endomorphism F has a meaningful relative characteristic polynomial over R; the scalar base action uI alone does not create Frobenius.

3.1 Young tableaux and a cyclic representation stratum

For a nonempty partition $\lambda \vdash m$, choose a numbering T of its boxes. On $V^{\otimes m}$ let $Y_T = R_T C_T$, the unnormalized row sum followed by the signed column sum. The classical identity $Y_T^2 = h_\lambda Y_T$, where h_λ is the hook product, makes Y_T/h_λ an idempotent in characteristic zero. Its image is $S_\lambda(V)$, one Schur copy, rather than the whole symmetric-group isotypic component [31, 32]. Its character is s_λ, computed by semistandard tableaux (weak rows, strict columns), and

$$\dim S_\lambda(k^d) = \prod_{(i,j)\in\lambda} \frac{d+j-i}{h_{ij}}.$$

Since $f^{\otimes m}$ commutes with the place permutations, it induces $S_\lambda(f)$ and respects composition and zero maps. Applying this at every vertex and arrow therefore gives a genuine representation of the monomial dessin algebra: $S_\lambda(N(b))S_\lambda(N(a)) = S_\lambda(0) = 0$ for a forbidden pair. The empty partition does not have this property: it sends zero arrow maps to the identity of k.

The Littlewood–Richardson rule gives natural intertwiners

$$S_\lambda(V) \otimes S_\mu(V) \cong \bigoplus_\nu S_\nu(V)^{\oplus c^\nu_{\lambda\mu}}.$$

Here $c^\nu_{\lambda\mu}$ counts semistandard tableaux of skew shape ν/λ and content μ whose row-reading word is Yamanouchi. Naturality makes the same decompositions intertwine all the arrows. The tensor operation is vertexwise with synchronous arrow maps, not $\otimes_A$ and not a Hopf-algebra tensor structure. There is no automatic tensor unit preserving these zero relations. These functors are not additive or exact; they cannot simply be applied degreewise to a projective resolution. They also need not preserve the fixed R-action on lattices.

In the elliptic theta set one of the two permitted three-cycle branches to zero. The other has arbitrary maps $X_1 : V_0 \to V_2$, $X_2 : V_2 \to V_1$, $X_3 : V_1 \to V_0$, with all spaces of dimension d. This is a closed representation stratum of our infinite dessin algebra. Put $M = X_3X_2X_1$.

Theorem 3.3 (Cyclic invariants and a local positive kernel)**.** *The invariant ring on this stratum is $k[c_1(M), \dots, c_d(M)]$, where c_j are characteristic coefficients. Its total-arrow-degree Hilbert series is $\prod_{j=1}^d (1-t^{3j})^{-1}$. Common multidegree (m,m,m) has basis $\chi_\lambda(M) = \operatorname{tr} S_\lambda(M)$, $\lambda \vdash m$, $\ell(\lambda) \le d$; other multidegrees have no invariants. Products obey the Littlewood–Richardson rule. Over $\mathbb{C}$, if $\|M_i\| \le R_0$ and $0 \le r$ and $rR_0^2 < 1$, the kernel*

$$K_r(M_i, M_j) = \det(I - rM_i \otimes \overline{M_j})^{-1} = \sum_\lambda r^{|\lambda|} \chi_\lambda(M_i)\overline{\chi_\lambda(M_j)}$$

is absolutely convergent and positive semidefinite.

Proof. Cycle-trace generation for quivers, including relation quotients, is the Le Bruyn–Procesi theorem [29, 30]. Every cycle here is a power of the full three-cycle, up to cyclic rotation. In characteristic zero Newton identities give the generators c_j. They are algebraically independent: take $X_1 = X_2 = I$ and arbitrary X_3. Scalar changes at the three vertices force equal arrow multidegrees. Restriction to that same section identifies degree (m, m, m) with degree-m symmetric polynomials in eigenvalues, whose Schur basis and product rule give the assertion. For the kernel use the ordinary Cauchy identity, first as a formal identity in eigenvalues. On Schur modules the operator norm of $S_\lambda(M)$ is at most $R_0^{|\lambda|}$, so its trace is bounded by $\dim S_\lambda(\mathbb{C}^d) R_0^{|\lambda|}$. The dimension Cauchy identity bounds the absolute series by $(1 - rR_0^2)^{-d^2}$. Each summand is a rank-one Gram kernel, proving positivity by absolute summation. □

The term for the empty partition is a constant scalar feature, not an allowed module-valued tensor unit. In dimension two,

$$(\operatorname{tr} M)^2 = \operatorname{tr} \operatorname{Sym}^2 M + \det M, \qquad \operatorname{tr} \operatorname{Sym}^2 M = \tfrac{1}{2}((\operatorname{tr} M)^2 + \operatorname{tr} M^2).$$

More generally $\det(I - zM) = \sum_j (-z)^j \operatorname{tr} \bigwedge^j M$ and its reciprocal is $\sum_m z^m \operatorname{tr} \operatorname{Sym}^m M$. These are actual determinant/Schur connections, not identifications with an arithmetic L-function. On continuous finite-length Λ-modules the monodromy is nilpotent and all positive-degree characters vanish. The displayed Cauchy positivity is local representation-theoretic positivity, not the global Weil form.

4 Lattice almost-split sequences and genuine diamond quotients

For the interval dessin let $X = (x)$, $Y = (y)$ in $\Lambda = k[[x, y]]/(xy)$, $R = k[[u]]$, $u = x + y$. A lattice is a finitely generated Λ-module free over R.

Theorem 4.1 (The nodal lattice calculation)**.** *Every lattice is $\Lambda^r \oplus X^a \oplus Y^b$. The almost-split sequences in this lattice category are*

$$0 \to Y \hookrightarrow \Lambda \xrightarrow{\times x} X \to 0, \qquad 0 \to X \hookrightarrow \Lambda \xrightarrow{\times y} Y \to 0.$$

In particular $\tau X = Y$, $\tau Y = X$. Multiplication by u between two copies of either row, followed by reduction modulo u, gives an actual nonzero exact three-by-three diagram.

Proof. Normalize M *inside its total quotient module*: $\overline{M} = BM$, $B = k[[x]] \times k[[y]]$. The conductor gives $u\overline{M} \subset M \subset \overline{M}$. Both branch projections of $W = M/u\overline{M} \subset k^p \oplus k^q$ are onto. Intersect W with the two factors; its remaining quotient identifies their two residual quotients. Separate bases for these pure factors and the common quotient put W into pure first lines, pure second lines and diagonal pairs. Lift branch bases over the DVRs. These are respectively the summands X, Y, Λ. Their endomorphism rings are local.

Exactness of the first sequence is $\operatorname{ann} x = Y$. It does not split: a map $X \to \Lambda$ lands in $\operatorname{ann} y = X$, and its composite with $\times x$ is a nonunit in $\operatorname{End} X = k[[x]]$. Every map $\Lambda \to X$ factors through this epimorphism, $\operatorname{Hom}(Y, X) = 0$, and every nonunit endomorphism of X is divisible by x and factors through $X \hookrightarrow \Lambda \to X$. Classification handles all finite sums; a unit X component would be a retraction. This is right almost splitness. The dual argument for nonsections out of Y proves left almost splitness. Endomorphisms fixing either structure map have constant term one, hence are units, proving minimality. Swap x, y for the other sequence. Finally all three terms are R-free. Multiplication by u is injective and reduction preserves exactness, proving the diagram assertion. □

The lattice AR quiver has arrows $Y \to \Lambda \to X$ and $X \to \Lambda \to Y$; the mesh composites are zero. Its projective vertex Λ is not the endpoint of another nonsplit AR sequence. AR meshes do not mean that all rows and columns of any drawn grid are short exact.

For a second, explicit family fix $n \geq 3$ and the rational Belyi map

$$\beta_n(z) = \frac{1}{1 - T_n(z)^2} = \frac{1}{(1 - z^2)U_{n-1}(z)^2},$$

where T_n, U_{n-1} are Chebyshev polynomials. It has passport $(2n), (2^n), (2^{n-1}, 1, 1)$. Its contracted dual graph is a path, identifying its genus-zero dessin. The associated infinite algebra has vertices $1, \ldots, n$, endpoint loops l, r, opposite arrows $a_i : i \to i + 1$, $b_i : i + 1 \to i$, and precisely the relations $l^2, r^2, a_i b_i, b_i a_i$. Its single permitted cycle is $l a_1 \cdots a_{n-1} r b_{n-1} \cdots b_1$; the order rank is $(2n)^2$.

Retain one arrow at each edge in an orientation Ω, and kill its opposite. The quotient is exactly the string algebra $H_n(\Omega)$ of Huang–Lin–Su [46]; the killed-arrow ideal contains J^{n+2}, so it is also an actual quotient of Λ_n. Inflation identifies its module category with the full abelian quotient-annihilator subcategory $\mathcal{C}_\Omega$ of Λ_n-modules. This is generally not extension closed in the whole order category. The imported AR property is intrinsic to $\mathcal{C}_\Omega$, not automatically an ambient lattice or finite-length AR property. The extended edition gives nine explicit string modules, all twelve maps and the four exact meshes from the supplied diamond resource, now situated in this actual dessin quotient.

The other supplied resources also have a precise home. For the interval order the subcategory annihilated by J^2 is $\operatorname{Mod} B_0$, $B_0 = \Lambda/J^2$. This is a domestic string quotient, not itself a gentle algebra. Inflation is full and preserves purity, and $\operatorname{Zg} B_0$ is a closed subspace of $\operatorname{Zg} \Lambda$. Ringel's infinite-word/pure-injective program and its proof by Puninski–Prest apply there [43, 44]; the isolated band xy^{-1} gives Krull–Gabriel and relative Cantor–Bendixson dimension two by [45, Theorem10.19]. This is not an ambient rank calculation for all Λ-modules. The parameter in $x = J_2, y = \lambda J_2$ survives even though all cycle traces vanish; the pp condition $\exists v\ (z = xv\ \wedge\ (y - \lambda x)v = 0)$ detects it. These modules are u-torsion and are not nonzero order lattices. Infinite-string and locally finite AR theorems must retain their distinct finiteness hypotheses; a surviving permitted cycle has infinitely many paths at a vertex.

5 Leavitt relations, the constrained Fock algebra and spectrum

Work now over $\mathbb{C}$. Ordinary Leavitt and graph C* constructions [47, 48] cannot simply impose our original forbidden products: if $ab = 0$ and CK1 says $a^*a = p_{t(a)}$, then $0 = a^*ab = b$. Every arrow is a forbidden successor, hence every arrow and then every vertex is zero. CK2 is not needed. The ordinary graph-Toeplitz CK1 setting has the same collapse.

The valid permitted-arrow-state graph instead consists of the disjoint directed σ-cycles. Its Leavitt algebra is $\bigoplus_c M_{|c|}(k[z, z^{-1}])$, and its graph C* algebra is $\bigoplus_c M_{|c|}(C(\mathbb{T}))$. These forget the α-gluing; both theta dessins give the same answer. They are not the completed original order. Nor is the ordinary Toeplitz algebra of a cycle of length $\ell > 1$ equal to $M_\ell(\mathcal{T})$: their K_0 ranks are ℓ and one respectively [50].

Let $\mathcal{H} = \ell^2(\mathcal{P})$ for all permitted finite paths, including vacua e_v. Define $L_a \delta_p = \delta_{ap}$ when the prefix is permitted and composable, and zero otherwise. Let P_v project onto paths starting at v, $Q_a = L_a L_a^*$ and v_v onto the vacuum. Then

$$L_a L_{\phi(a)} = 0, \quad L_a^* L_a = P_{t(a)} - Q_{\phi(a)} = v_{t(a)} + Q_{\tau(a)}, \quad P_v - \sum_{s(a)=v} Q_a = v_v.$$

These follow on basis vectors. The different initial projection avoids CK1 collapse. Write $\mathcal{T}_A = C^*(P_v, L_a)$, a constrained-path Fock C* algebra in the monomial/subproduct setting [51, 52]. The polynomial A acts faithfully: applying a finite path sum to the sum of vacua recovers its coefficients. This representation need not extend to all formal Λ series: $\sum_{m\geq 0} c^m$ along a cycle would send its vacuum to a nonsquare-summable sum.

Theorem 5.1 (Compact extension and incidence index)**.** *Let $\mathcal{H}_v$ be the terminal-path sector. There is an exact sequence*

$$0 \longrightarrow \bigoplus_{v\in Q_0} \mathcal{K}(\mathcal{H}_v) \longrightarrow \mathcal{T}_A \longrightarrow \bigoplus_{c\in \mathrm{cyc}(\sigma)} M_{|c|}(C(\mathbb{T})) \longrightarrow 0.$$

With $\delta[u] = [1 - W^*W] - [1 - WW^*]$, its boundary map is $\delta(c) = -\sum_{h\in c} e_{[h]}$, minus the unsigned incidence map B_Γ of the cellular graph. Thus, for its E edges,

	$K_0(\mathcal{T}_A)$	$K_1(\mathcal{T}_A)$
Γ *bipartite*	$\mathbb{Z}^{E+1}$	$\mathbb{Z}$
Γ *nonbipartite*	$\mathbb{Z}^E \oplus \mathbb{Z}/2$	0.

Proof. All prefixes preserve the terminal sectors. For paths p, q ending at v, $L_p v_v L_q^* = |\delta_p\rangle\langle\delta_q|$. Therefore the vacuum ideal is exactly the displayed sum of compacts: it contains these matrix units and that sum is itself an ideal. In the quotient, L_a has initial projection $Q_{\tau(a)}$ and range Q_a, a graph CK family for the permitted-state cycles. The Q_a have infinite rank and nonzero quotient images; the length gauge action descends. Gauge-invariant uniqueness [48, Theorem2.1] identifies the quotient as claimed.

For a cycle c based at arrow a, lift its corner unitary by $W = L_c Q_a + (1 - Q_a)$. Then $W^*W = 1$ and $1 - WW^* = Q_a - L_c Q_a L_c^*$ projects onto the first $|c|$ paths starting with a. Their terminal vertices count exactly its incident darts, including twice for a loop. This proves the index formula. The six-term exact sequence gives $K_1 = \ker B_\Gamma$ and $0 \to \operatorname{coker} B_\Gamma \to K_0 \to \mathbb{Z}^{\#\,\mathrm{cyc}\,\sigma} \to 0$. The free last term splits abstractly. Spanning-tree integer elimination gives Smith entries 1 repeated $V - 1$ times and one zero in the bipartite case; an odd cycle instead adds the entry two. This proves the groups. □

The nodal interval has $K_0 = \mathbb{Z}^2$, $K_1 = \mathbb{Z}$; the single-loop sphere has $\mathbb{Z} \oplus \mathbb{Z}/2$, zero. Both theta dessins have $\mathbb{Z}^4$, $\mathbb{Z}$. Coarse K-theory does not recover genus or the arithmetic factors.

Proposition 5.2 (The degree spectral triple)**.** *Define $N\delta_p = |p|\delta_p$ on $\{\xi : \sum_p |p|^2|\xi_p|^2 < \infty\}$. With the polynomial creation/annihilation *-algebra this is an ungraded spectral triple, p-summable exactly for $p > 1$. With N_+^{-s} zero on the kernel,*

$$\operatorname{Tr} e^{-tN} = E + \frac{2E}{e^t - 1}, \quad \operatorname{Tr} N_+^{-s} = 2E\zeta(s), \quad \operatorname{Tr}\big(\chi(N)N_+^{-s}\big) = 2EL(s,\chi)$$

for $t > 0$ and $\operatorname{Re} s > 1$ in the last two formulas, where χ is an externally chosen Dirichlet character.

Proof. There are E vacua and $2E$ paths of every positive length: each starting arrow has a unique continuation. The real diagonal operator is self-adjoint on the stated maximal domain; finite supports are a core and its resolvent is compact. Prefixes preserve the domain and $[N, L_a] = L_a$, $[N, L_a^*] = -L_a^*$, so polynomial commutators are bounded. The trace formulas and summability reduce to the corresponding scalar series. The character multiplier is bounded; membership in $\mathcal{T}_A$ is not asserted. □

Length parity anticommutes with the creators but commutes with N, so it is not an even-spectral-triple grading. The ordinary odd Fredholm class of this positive N is trivial: $N(1 + N^2)^{-1/2} - I$ is compact. The spectrum depends only on E; zeros of the scalar zeta trace are not eigenvalues of N.

6 Arithmetic attachment, descent and the remaining comparison

The compact ideal gives a precise home for an arithmetic determinant, with no claim that the arithmetic is derived from the dessin. Fix one terminal sector. Its two paths of each length $n \geq 1$ give orthonormal vectors $\xi_n = (\delta_{p_{n,1}} + \delta_{p_{n,2}})/\sqrt{2}$ and an isometry $U : \ell^2(\mathbb{N}_{\geq 1}) \to \mathcal{H}_v$.

For a nonprincipal character χ put $A_n = \sum_{j \leq n} \chi(j)$, $B_0 = \sup |A_n| \leq q_\chi$, and, on $2^k \leq n < 2^{k+1}$, set

$$w_n = \sqrt{n}(k+1), \quad u_n = A_n/w_n, \quad v_n(s) = w_n(n^{-s} - (n+1)^{-s}), \quad \varphi_s(z) = \sum_n v_n(s) z_n - z_1.$$

Here $u_1 = 1$. For $\operatorname{Re} s > 0$, $u, v(s) \in \ell^2$ and the pairing is complex linear, with no conjugation of s. Define

$$\widetilde{K_s} = U(u \otimes \varphi_s)U^*, \qquad \det_F(I + \widetilde{K_s}) = L(s, \chi).$$

Indeed dyadic blocks give $\|u\|^2 \leq 2B_0^2$, and on a compact set $\operatorname{Re} s \geq \sigma_0 > 0$, $|s| \leq M$, the squared v tail is bounded by $M^2 \sum_{k \geq J} (k+1)^2 2^{-2\sigma_0 k}$. Thus the rank-one family is trace-norm holomorphic. Its determinant is $1 + \varphi_s(u)$, equal by Abel summation to L. Replacing v by $P_N v$ and adding $A_N(N+1)^{-s} e_1^*$ to the functional gives exactly every raw partial sum S_N, with compact-uniform trace-norm convergence; $\|UXU^*\|_1 = \|X\|_1$. For primitive χ, completion replaces the functional by $g(s) \sum v_n(s) z_n - z_1$, with the standard nonzero gamma factor g, rather than multiplying the whole old functional. The extended edition supplies the quantitative estimates.

This inserts a scalar encoding into one actual dessin compact ideal; it is not known to be A-linear, a derived chain map or Frobenius compatible. A common fixed positive metric cannot make this same fixed-range family symmetric for all critical heights: symmetry forces its functionals to be proportional, whereas the coefficient ratio at heights $0, t \neq 0$ is asymptotic to $(\sigma + it)n^{-it}/\sigma$, with two opposite subsequential limits. Different positive realizations remain possible.

Proposition 6.1 (Complex dessin data do not determine arithmetic descent)**.** *The rational Belyi pairs $E_d : y^2 = x^3 - d$, $\beta_d = x^3/d$, for $d = 1, 2$, are isomorphic over $\mathbb{C}$ but have distinct Frobenius polynomials at the good prime seven: $1 - 4T + 7T^2$ and $1 - T + 7T^2$.*

Proof. Choose $a^3 = d$, $b^2 = d$. The projective isomorphism $(x, y) \mapsto (ax, by)$ from E_1 to E_d preserves the Belyi map. The degree-six clean passports are the elliptic theta passport already computed. Directly counting y for $x = 0, \ldots, 6$ gives respectively $(0, 1, 1, 0, 1, 0, 0)$ and $(0, 0, 0, 2, 0, 2, 2)$. Including infinity yields four and seven points, hence $a_7 = 8 - \#E_d(\mathbb{F}_7) = 4, 1$. Both discriminants $-432d^2$ are nonzero modulo seven. □

Any invariant of the complex dessin or its undecorated order/Fock algebra alone is the same for these two inputs, so cannot recover both factors without descent data. This does not obstruct a theory including a specified rational model and compatible Galois action. Elliptic H^1 has weight one and unshifted center $s = 1$, unlike Dirichlet center $1/2$; the example is not a counterexample to GRH.

The recent NCG literature supplies concrete comparison targets, with strong hypotheses. Connes–Consani construct finite abelian covers of the adelic Riemann sector with prime-orbit length $\log p$ and arithmetic Frobenius monodromy [53, Theorem3.12]. The prolate/local-factor and moment constructions [54, 55] give particular finite-prime analytic models, not a uniform positive comparison for all completed L-functions. The zeta-spectral-triple determinant theorem [56, Theorem5.10] assumes a specified restricted Weil form, simple even bottom eigenfunction and boundary normalization. Its infinite comparison is still an additional step. The distributional real-zero theorem [57, Theorem6.1]

concerns a particular interval quadratic form, essential self-adjointness on its trigonometric core and an even simple ground state; a Fock vacuum alone verifies none of those arithmetic identifications. Deninger's periodic-packet construction and real-cohomology obstruction [60, 62] further specify, rather than remove, these issues. Full theorem-level reading coverage and exact versions are recorded alongside this draft.

A complete approach must construct arithmetic descent and prime actions on actual order/derived or analytic objects, prove determinant-line and local-factor compatibility, control regularization and conductor/gamma normalization, and establish a positive realization for that *same* function. Finite-field cohomological trace formulas and Deligne purity supply the classical model [20, 21]; alternating duality alone is not positivity or purity. In a number-field construction these are obligations, not imported conclusions.

For a sharply testable final analytic target, put $F_\chi(z) = \Lambda(1/2 + iz, \chi)$. Suppose entire functions $h_m(z)\det(zI - H_m)$, with H_m finite self-adjoint and h_m entire nowhere zero, converge locally uniformly on $\mathbb{C}$ to this nonzero completed function. Hurwitz on every disk disjoint from $\mathbb{R}$ then proves all its zeros real, hence the Dirichlet critical-line assertion. Establishing this convergence with independently constructed H_m is exactly the missing load-bearing claim; a rank-one encoding or a reformulation does not establish it. Hecke, Dedekind and Artin extensions must state their analytic scope anew.

7 Verification and research frontier

The new Lean files construct the node as an actual subring of $k[[X]] \times k[[X]]$ with equal constants, prove its annihilators, regular common parameter, augmentation, alternating projective resolution and a bijective pair of power-series coordinates. A second file proves CK1 collapse under the stated permutation/coverage hypotheses, determinant base change, actual 2×2 symmetric-square functoriality and relation preservation, a Schur character identity and the fixed-parameter lattice failure. These thirty printed theorem types have only standard Lean foundations. They do not formalize general dessins, abstract Ext/AR/Schur–Weyl, C* exactness, Fredholm determinants, arithmetic descent or GRH. Earlier formal layers remain preserved with their literal scopes.

Exact-rational diagnostics independently check the displayed dessins, path identities, Schur and Young examples and the nine-object grid. These finite checks supplement the proofs, rather than prove infinite spectral or arithmetic assertions. The next arithmetic target is an exact descent/Frobenius-compatible construction on the labelled order/Fock data, with a determinant-preserving positive comparison. The local Schur kernel and the simple degree spectrum alone do not supply it. All source clones and earlier research are retained; the earlier general focused PDFs are archived at the preceding commit.

A Expanded dessin, resolution and derived calculations

A.1 The cycle order in coordinates

Number a permitted cycle in travel order $0, \dots, \ell - 1$. Map an arrow $i \to i+1$ to $E_{i,i+1}$ for $i < \ell - 1$, and the final arrow to $u_c E_{\ell-1,0}$. A path from i to j then gives $u_c^r E_{ij}$, where r counts cut crossings. These are linearly independent path normal forms, and every positive-length permitted path has exactly one such expression on exactly one branch. The full-cycle sum in a branch is $u_c I$. At a paired edge its two diagonal occurrences come from a single original idempotent; no positive-length coefficients are identified. This proves the order description including self-pairings within one cycle.

Although the suborder is generically a product of full matrix algebras, it is not their unglued product at the special fiber.

For the interval this becomes the literal fiber product

$$\Lambda = \{(f,g) \in k[[X]]^2 : f(0) = g(0)\}, \qquad x = (X,0),\ y = (0,X),\ u = (X,X).$$

Every element has a unique expression $a(u) + b(u)x$: take $a = g$ and $b = (f-g)/X$. Thus it is R-free on $1, x$ with $x^2 = ux$ and $y = u - x$. The annihilator of x consists of pairs $(0,g)$ with $g(0) = 0$, exactly $y\Lambda$; the other branch is symmetric. Multiplication by u is injective, coefficientwise. These are actual formal power-series statements, not conclusions from finite truncations.

A.2 The two-arm complex and its calculations

For an arrow a write $f(a)$ for its unique forbidden successor. The summand in $P_n(i)$ indexed by the initial arrow a has endpoint $t(f^{n-1}(a))$. Its map multiplies by that last arrow. Thus the resolution can be computed by iterating a finite permutation, while the projectives themselves remain infinite-dimensional. The Hom complex against a module N is the direct sum of the endpoint spaces with differentials given by the same arrow matrices. Tensor against a left module instead uses the same path entries acting on the left. No rank or exactness claim for an arbitrary N is inferred by counting endpoints alone.

For the nodal residue field the maps are particularly explicit:

$$\cdots \xrightarrow{\mathrm{diag}(x,y)} \Lambda^2 \xrightarrow{\mathrm{diag}(y,x)} \Lambda^2 \xrightarrow{(x \quad y)} \Lambda \longrightarrow k \longrightarrow 0.$$

If $xz_1 + yz_2 = 0$, its first branch says the first branch of z_1 is zero, and its second branch says the second branch of z_2 is zero. Hence $z_1 = yw_1$, $z_2 = xw_2$. The same annihilator calculation proves the alternating kernels in all degrees. Projectivity is genuine: the terms are regular modules or their finite direct sums. It is not a finite free resolution, despite the period of its positive tail.

For the elliptic theta the quiver vertices are $v_0 = \{0,3\}$, $v_1 = \{1,4\}$, $v_2 = \{2,5\}$. The arrows a_0, a_3 go $v_0 \to v_2$, a_2, a_5 go $v_2 \to v_1$, and a_1, a_4 go $v_1 \to v_0$. The two forbidden arms from v_0 are

$$a_0, a_5, a_1, a_3, a_2, a_4, \ldots, \qquad a_3, a_2, a_4, a_0, a_5, a_1, \ldots.$$

Their top pairs are $(v_2,v_2), (v_1,v_1), (v_0,v_0)$, repeated. The three steps exchange the arms. Their distinct top pairs exclude shorter unordered positive periods. The spherical theta instead has forbidden two-cycles and successive top pairs $(v_2,v_1), (v_0,v_0)$, giving period two. Thus even identical hereditary overorders need not give identical resolutions.

Quadratic Koszul duality here has a directly checkable local meaning: relations complement the two possible successors. The dual dessin replaces the graph rotation by ϕ^{-1}. In the Ext algebra against A/J, concatenating the forbidden-word basis gives the product when the join is forbidden; a permitted join gives zero. One can see this by lifting a length-r basis class to the resolution: discard the prescribed initial r letters and identify the remaining summand with its suffix resolution, taking zero on other prefixes. These prefix-discard comparison maps commute with the suffix differentials, with the standard shift signs when viewed as shifted complexes. The concatenated indexing word exists exactly when its join is forbidden; evaluation of the two basis classes on that word gives one. If the join is permitted there is no such indexing word and the product is zero. Composition therefore concatenates indexing words, with the chosen right-module/Yoneda opposite convention. This is a graded algebra statement; it neither makes the unbounded resolution perfect nor constructs an equivalence of all formal-completion derived categories.

For an arbitrary bounded projective complex $P^\bullet$, its determinant line over a commutative coefficient ring is $\bigotimes_n (\det P^n)^{(-1)^n}$, with determinant duals for negative exponents. A chain automorphism acts on this line; a quasi-isomorphism gives the corresponding determinant-functor isomorphism. For a general endomorphism F, the expression for $1 - zF$ uses the locus where it is invertible and gives a rational determinant-line action/trivialization, not an everywhere invertible line map at singular parameters. Additivity is compatible with exact triangles under the usual determinant-functor conventions [17, 18]. In our order context an R-linear determinant uses the finite free underlying R-modules; it must specify a separate endomorphism. Determinants of a differential presentation, alternating determinants of a chain action, regularized analytic determinants and the Cartan path matrix are four distinct constructions. Their equality requires a proof, not their shared name.

The geometric model literature [6, 8, 9] is useful for identifying which crossings and gradings would have to be represented by a functor. To apply it here one must specify the category of polynomial strings, completed lattices or finite-length quotients; construct its surface/curve objects; and prove that degree-r intersections compute the Hom group into shift r, with local systems and boundary exceptions included. We have supplied explicit walks and maps in the examples, not claimed this general completed-order theorem. On a finite word, a map can already be computed rigorously without such a theorem: assign a coefficient matrix between equal-vertex occurrences and solve $F_{t(a)} N(a) = N'(a) F_{s(a)}$ for every arrow. The diamond calculations below implement exactly this system.

B Semi-invariants, tableaux and explicit intertwiners

B.1 Derived information retained by a presentation

Let $K = \operatorname{im} f$. The injection $\operatorname{Hom}(K, N) \to \operatorname{Hom}(P_1, N)$ descends to the Ext injection used in Theorem 3.1; its kernel modulo $\operatorname{im}\operatorname{Hom}(P_0, N)$ is zero because the map $P_1 \to K$ is onto. The missing part of the cokernel detects maps on P_1 not descending through K. This explains exactly why an injective resolution map is needed for its full Euler interpretation.

In the node example the resolution of $M = \Lambda/(x)$ has maps $x, y, x, y, \ldots$. On $N = k[y]/(y^2)$ the Hom cochain maps are $0, y, 0, y, \ldots$. Thus $\operatorname{Hom}(M, N) = N$ and $\operatorname{Ext}^1(M, N) = \ker y$ have dimensions two and one; the cokernel of the first zero map is two-dimensional. In the elliptic-square example the first four Hom maps are $1, 0, 0, 0$, periodically. Hence degrees zero and one vanish, degrees two and three are k, and degree four vanishes. This also makes clear why a nonzero scalar semi-invariant does not give a derived orthogonality theorem.

With coordinate action $(gF)(N) = F(g^{-1}N)$ the Schofield weight is $a - b$, the negative of the point-evaluation weight. For a single arrow it is the same elementary distinction as the action on $\operatorname{Hom}(V, W)$ versus on its coordinate functions $V \otimes W^*$. The Cauchy decomposition for coordinate functions is

$$\operatorname{Sym}^m(V \otimes W^*) \cong \bigoplus_{\lambda \vdash m} S_\lambda(V) \otimes S_\lambda(W)^*.$$

A relation quotient can kill these components; its ambient multiplicities do not descend unchanged. Acyclic-quiver semi-invariant/LR results such as [35] retain their acyclicity assumptions. The explicit two-arrow slice below is acyclic, whereas the whole theta quiver is not.

B.2 One Young operator and one LR decomposition in coordinates

For $\lambda = (2,1)$ number the first row $1,2$ and second row 3. Then

$$Y = (I + P_{12})(I - P_{13}), \quad Y^2 = 3Y, \quad \dim \operatorname{im}(Y \mid (k^2)^{\otimes 3}) = 2.$$

The hook product is three; its Specht dimension is $3!/3 = 2$, also two here for a different reason. The normalized projector extracts one $S_{(2,1)}$ copy. The trace $\operatorname{tr}((Y/3)M^{\otimes 3})$ is therefore the Schur character, not twice that character. This fixes the common row/column normalization ambiguity.

Write $t = \operatorname{tr} M$, $\Delta = \det M$ for a two-dimensional endomorphism. Its complete characters obey $h_0 = 1$, $h_1 = t$, $h_m = t h_{m-1} - \Delta h_{m-2}$, and

$$\chi_{(a,b)}(M) = \Delta^b h_{a-b}(M), \quad a \geq b \geq 0.$$

In particular $h_2 = t^2 - \Delta$, $h_3 = t^3 - 2t\Delta$, and $\chi_{(2,1)} = t\Delta$. Thus $t h_2 = h_3 + \chi_{(2,1)}$ is the LR identity $V \otimes \operatorname{Sym}^2 V \cong \operatorname{Sym}^3 V \oplus S_{(2,1)}V$, with $S_{(2,1)}V \cong \det V \otimes V$.

An explicit natural splitting avoids relying just on characters. In characteristic zero multiplication $m : V \otimes \operatorname{Sym}^2 V \to \operatorname{Sym}^3 V$ has natural section

$$s(v_1 v_2 v_3) = \tfrac{1}{3} \sum_{i=1}^{3} v_i \otimes \prod_{j \neq i} v_j.$$

The map $j : \bigwedge^2 V \otimes V \to V \otimes \operatorname{Sym}^2 V$ sends $(v \wedge w) \otimes z$ to $v \otimes wz - w \otimes vz$. In dimension two it is injective: for a basis e_1, e_2 its images at $z = e_1, e_2$ have distinct independent coordinates. They lie in $\ker m$, whose dimension is $6 - 4 = 2$. Hence (s, j) is the required isomorphism. Tensorial definitions make both maps commute with every linear endomorphism, including singular arrow maps. This is a genuine LR intertwiner on our dessin representation stratum.

For a concrete theta representation take

$$X_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad X_2 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad X_3 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Then $M = X_3 X_2 X_1 = \left(\begin{smallmatrix} 2 & 2 \\ 2 & 3 \end{smallmatrix}\right)$, $t = 5$, $\Delta = 2$, $h_2 = 23$, $h_3 = 105$ and $\chi_{(2,1)} = 10$. The identity is $115 = 105 + 10$. Since the other permitted branch is zero, every forbidden product is zero; this is an actual representation of the infinite elliptic dessin algebra.

In the invertible-arrow open stratum one can gauge X_1, X_2 to the identity, leaving conjugation of M. The invariant-ring proof in the core works also at singular arrows by cycle-trace generation, not by assuming that this gauge choice extends there. The degree-one monodromy coefficients have total arrow degree three; this is why the Hilbert-series denominator uses $3j$, not j.

For the Cauchy kernel choose the Schur subspace inside the tensor power with its inherited Hilbert norm. $M^{\otimes m}$ preserves it and has norm at most $\|M\|^m$. This proves the trace bound for nonnormal matrices as well as diagonal ones. Formal Cauchy identities extend by polynomial continuity in matrix entries; the stated absolute bound then justifies analytic summation. Gram positivity does not assert entrywise positivity or positivity of an unrelated arithmetic form. For three exact test matrices M, $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 2 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 1 \end{smallmatrix}\right)$, the choice $r = 1/100$ satisfies $r R_0^2 \leq 21/100 < 1$ using Frobenius norms, and all seven principal minors of the rational kernel matrix are nonnegative.

B.3 Two precise functorial limitations

At any actual vertex simple S of A, the split exact sequence $0 \to S \to S \oplus S \to S \to 0$ uses first inclusion and second projection. Termwise Sym^2 has dimensions $1, 3, 1$ at that vertex; its middle kernel contains the mixed monomial as well as the first square, but its first image contains only the first square. Middle homology is S. Thus even split exactness is not preserved. Genuine derived Schur constructions instead use mixed tensor degrees, signed total differentials and suitable flat/projective replacements [32, 33]; their membership in a particular order category must still be checked.

The R-lattice $M = Re_x \oplus Re_y$ with $x = \mathrm{diag}(u, 0)$, $y = \mathrm{diag}(0, u)$ has $x + y = uI$. On $\mathrm{Sym}^2_R M$ the images are $\mathrm{diag}(u^2, 0, 0)$ and $\mathrm{diag}(0, 0, u^2)$. Their sum kills the mixed vector, hence equals neither uI nor $u^2 I$. This is a specific failure to retain the original central action, not a failure of the polynomial monomial representation functor. Even a common ramified base substitution does not repair this mixed vector.

Semi-invariants retain useful moduli on nilpotent modules. In the theta quiver keep only $a_0, a_3 : v_0 \to v_2$, put their scalar values $(x, y) = (1, \lambda)$ and all other arrows zero. The v_1 simple splits off. Every length-two path acts by zero, so these are continuous completed modules. The two scalars have the same weight g_{v_2}/g_{v_0}, and y/x is a rational invariant on $x \neq 0$, giving the projective pencil $[x : y]$. It is not a regular invariant on the whole affine locus, whose orbit closures meet zero.

For the interval quotient take $x = J_2$, $y = \lambda J_2$. With its top and socle specified, the separated Kronecker coordinates are again $(1, \lambda)$; the relation to the same-weight ratio is exact. The pp formula in the core is stronger than merely noticing a ratio: on M_λ its solution set is $\mathrm{im}\, x$, whereas on M_μ, $\mu \neq \lambda$, the equation $(\mu - \lambda)xv = 0$ forces $z = xv = 0$. Different parameters are therefore distinguishable by actual positive-primitive conditions even though every cycle trace is zero.

C Full lattice and diamond diagrams

C.1 The nodal exact diagram

The normalization proof of Theorem 4.1 uses BM inside the generic module. Replacing it without justification by $B \otimes_\Lambda M$ could introduce torsion and would not prove the same statement. Once the subdirect residue subspace is decomposed, the branch basis changes lift over complete DVRs; taking inverse images modulo the conductor gives the direct decomposition, not just a decomposition of residues.

Here is the promised diagram, with the terminal short exact column for every position:

$$\begin{array}{ccccccccc}
0 & \longrightarrow & Y & \hookrightarrow & \Lambda & \xrightarrow{x} & X & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle u} & & \downarrow{\scriptstyle u} & & \\
0 & \longrightarrow & Y & \hookrightarrow & \Lambda & \xrightarrow{x} & X & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & Y/uY & \hookrightarrow & \Lambda/u\Lambda & \longrightarrow & X/uX & \longrightarrow & 0
\end{array}$$

Each vertical column is completed by zero above and below. In the bottom row identify $Y/uY = k$ via generator y, $X/uX = k$ via x, and $\Lambda/u\Lambda = k[\epsilon]/(\epsilon^2)$ with $x = \epsilon$, $y = -\epsilon$. Its first map is $1 \mapsto -\epsilon$, its second the augmentation. It is nonsplit as a dual-number sequence. This explicit order reduction, not an unrelated finite algebra, explains its appearance.

For a further Ext calculation, resolve $X \cong \Lambda/(y)$ with successive maps $y, x, \ldots$. On Y the first is multiplication by y and the second zero, so $\mathrm{Ext}^1(X,Y) = Y/yY = k$; on X the first is zero and the second injective, so $\mathrm{Ext}^1(X,X) = 0$. Interchanging branches gives the other two cases. The two nonsplit lattice rows realize the nonzero cross classes. Their Hom spaces are finite over R, generally infinite over k.

C.2 Chebyshev geometry and the exact quotient

The identity $1 - T_n(z)^2 = (1-z^2)U_{n-1}(z)^2$ follows from $z = \cos\theta$ and the sine/cosine multiple-angle formulas, hence as a polynomial identity. $T_n' = nU_{n-1}$. Therefore β_n has its only zero at infinity, of order $2n$; its n points over one are double since $\beta_n - 1 = T_n^2/(1-T_n^2)$; and its poles are simple at ± 1 and double at the $n-1$ interior roots of U_{n-1}. The derivative has no additional critical values. The total ramification is $(2n-1) + n + (n-1) = 4n-2$, as required for a degree-$2n$ rational Belyi map. The dual graph has $n+1$ vertices of degrees $1, 1, 2, \ldots, 2$, and n edges; it is a connected path. Its planar ribbon embedding is unique, so this identifies the particular nested-loop dessin, not only its passport.

In the quotient retaining orientation Ω, any surviving path traverses each chain edge at most once and either endpoint loop at most once. Its length is at most $n+1$; thus the killed-arrow ideal contains J^{n+2}. The quotient of A_n and of its completion is the same $H_n(\Omega)$. For $n = 3$ with forward chain arrows, its path counts in lengths zero through four are $3, 4, 3, 2, 1$, total thirteen. These counts describe a quotient representation category of the infinite order, not a change of primary object.

Inflation is full because the two algebras have exactly the same actions on these annihilated objects. It preserves internal exact sequences. It does not preserve arbitrary ambient extension closure: the two adjacent simples are annihilated by the killed reverse arrow, but the two-dimensional extension having only that arrow nonzero is not. Likewise endpoint-local freeness over $k[l]/l^2$ or $k[r]/r^2$ in [46] is not R-freeness. To transfer an almost-split property to a larger category one must prove the requisite approximation/factorization statements; locally finite quiver criteria [40, 41, 42] do not apply just because each vertex has two arrows. The permitted cycles violate interval/Hom finiteness.

C.3 Nine actual strings and all maps

Take $n = 3$ with $1 \xrightarrow{a} 2 \xrightarrow{b} 3$, loops l, r and $l^2 = r^2 = 0$ in the quotient. Here b denotes the retained forward a_2, not a reverse arrow of the original doubled chain. The following words are in travel order. A superscript $-$ means a formal inverse in a word, not an inverse algebra element.

Object	word	start	dimension vector
A	$l^- abrb^- a^- l^-$	1	$(4,2,2)$
B	$l^- ab$	1	$(2,1,1)$
C	$l^- abr^-$	1	$(2,1,2)$
D	$abrb^- a^- l^-$	1	$(3,2,2)$
E	ab	1	$(1,1,1)$
F	abr^-	1	$(1,1,2)$
G	$r^- b^- a^- labrb^- a^- l^-$	3	$(4,3,4)$
H	$r^- b^- a^- lab$	3	$(2,2,3)$
I	$r^- b^- a^- labr^-$	3	$(2,2,4)$

A basis vector is assigned to each visited position $0, \ldots, \ell(w)$ at its vertex. A direct letter sends its preceding vector to its following vector; an inverse letter sends the following vector to the preceding

vector. Sum these elementary maps for each arrow and put all unmentioned entries zero. This defines every matrix, including repeated loops, and verifies the relations. Set the killed reverse arrows to zero to obtain actual Λ_3-modules.

Their signed grid is

$$\begin{array}{ccccc} A & \xrightarrow{p} & B & \xrightarrow{i} & C \\ {\scriptstyle -p}\downarrow & & \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle -p} \\ D & \xrightarrow{p} & E & \xrightarrow{i} & F \\ {\scriptstyle -i}\downarrow & & \downarrow{\scriptstyle i} & & \downarrow{\scriptstyle -i} \\ G & \xrightarrow{p} & H & \xrightarrow{i} & I. \end{array}$$

Horizontal projections keep the initial word positions and kill the remaining positions. Horizontal inclusions append the final r^-, keeping old positions. Upper vertical projections delete the initial l^-: old position $j \geq 1$ maps to position $j-1$, and old position zero maps to zero. Lower vertical inclusions prepend $r^-b^-a^-l$: position j maps to $j+4$. Multiply these maps by the indicated signs. For every arrow these formulas commute with the string action, including the two positions adjacent to a cut or join; those are exactly where the loop inverse and chosen orientation matter.

Every small square has two length-two composites whose sum is zero. The exact sequences are

$$0 \to A \to B \oplus D \to E \to 0, \qquad 0 \to B \to C \oplus E \to F \to 0,$$
$$0 \to D \to E \oplus G \to H \to 0, \qquad 0 \to E \to F \oplus H \to I \to 0.$$

For example the maps out of A retain its initial positions $0,1,2,3$ in B and delete its first position in D, so their joint kernel is zero. The maps to E are onto, and the dimension vectors give $\dim A + \dim E = \dim B + \dim D$ vertexwise. The zero composite plus this equality proves exactness. The other three use precisely the same keep/delete or append/prepend rules; their vertexwise dimension equalities and jointly injective/surjective maps prove exactness. The shipped exact checker constructs all matrices and verifies these assertions independently over $\mathbb{Q}$.

The stronger almost-split property is imported by identifying $E = w = ab$ as the minimal string of type $(2,2)$ and these exact hook/cohook words with [46, Proposition3.12], using that paper's Propositions2.6/2.9 and Lemma3.10. Opposite-module conventions are reconciled by reversing travel/composition conventions; the closed doubled quiver is isomorphic to its opposite by exchanging the two arrows at each chain edge. Thus intrinsically $\tau E = A$, $\tau F = B$, $\tau H = D$, $\tau I = E$. The rows and columns of this grid are *not* generally short exact, unlike the nodal diagram. Two small cautions in that source are not needed here: a tube-rank label in Example3.9(3) is inconsistent with its listed mouth, and a general homogeneous-tube statement must not be read as a converse classifying every such object as a band. Neither is used to import these four meshes.

C.4 The pure-injective quotient and what its dimension means

For $B_0 = \Lambda/J^2 = k\langle x,y\rangle/(x,y)^2$, words alternate direct and inverse letters because every direct/direct and inverse/inverse adjacent pair is forbidden. Up to rotation and inverse its only primitive band is xy^{-1}; the bridge quiver has two oriented band vertices with no connecting nontrivial bridge. The longest path has length zero. The domestic string results [44, 45] therefore give relative dimension $0+2=2$, not an automatic dimension three for every one-domestic algebra. Finite-dimensional points have relative CB rank zero; the appropriate Prüfer/adic and one-sided periodic string points have rank one; the unique generic point has rank two. These statements concern the quotient category with its inherited pp structure.

Indeed a pp formula is a finite system of linear equations with existentially quantified unknowns. On a module annihilated by J^2 every coefficient reduces to B_0, so purity and pure-injectivity are preserved in both directions under inflation. The condition $MJ^2 = 0$ is a definable closed condition, giving the closed Ziegler subspace. Its relative ranks need not be ranks in the entire Λ spectrum. This subcategory is not extension closed: $\Lambda/(x^3, y)$ gives an extension of objects killed by J^2 that is not itself killed by J^2. Since u^2 annihilates every object of the subcategory, it contains no nonzero R-free lattice. Ringel's representation-theoretic node construction is also not a Morita equivalence identifying the geometric nodal order with the Kronecker algebra. Infinite-dimensional polynomial string classifications [38] retain finitely-controlled or pointwise-artinian assumptions; an arbitrary complete lattice does not automatically satisfy them as an algebraic A-module.

D The operator extension in detail

D.1 A corrected cycle lift and the integer boundary map

In the permitted-state graph each length-ℓ cycle has a full matrix-corner Laurent/circle coordinate. Its chosen K_1 generator is the full cycle at one state plus the complementary identity. In Fock space $L_c^* L_c = Q_a + v_{s(a)}$. Thus the tempting lift $L_c + (1 - Q_a)$ has overlapping initial pieces and is not an isometry. The corrected lift in Theorem 5.1 first restricts to Q_a:

$$W = L_c Q_a + (1 - Q_a), \quad W^* W = I, \quad I - WW^* = Q_a - L_c Q_a L_c^*.$$

A path starting with a and length greater than ℓ has uniquely the full prefix c followed by another path starting with a. The missing range consists exactly of lengths $1, \dots, \ell$. Its terminal multiset is the original graph's incident-edge multiset at that cycle vertex. These finite defect vectors all lie in terminal compact ideals; their classes give the claimed unsigned incidence entries. The sign convention affects only the overall sign of the boundary map.

Here is the Smith calculation without spectral assumptions. Choose a spanning tree of the connected underlying graph and one root. By changing signs on one bipartition of the tree, its $E \times V$ incidence submatrix becomes the ordinary oriented tree incidence, with $V - 1$ unit pivots. After eliminating these pivots, a remaining column represents the common root coordinate, and each nontree edge evaluates on it as zero when its fundamental cycle is even, or ± 2 when odd. A loop is the latter case. If all are even the graph is bipartite and this column is zero; otherwise its entries have gcd two, giving the final pivot two. Remaining rows are free. Consequently the kernel/cokernel, then the six-term sequence, give the K groups stated in the core. No theorem reconstructing a cyclic embedding from those groups is asserted.

The compact ideal is block diagonal, not $\mathcal{K}(\mathcal{H})$: prefixes never move terminal vertices, so cross-terminal rank-one maps do not belong to the displayed ideal. This is why the arithmetic insertion must choose one terminal sector. A graph/Cuntz boundary quotient kills the vacua and hence this ideal, so that insertion need not survive passage to the quotient.

D.2 Spectral domain and arithmetic trace

A vector belongs to the adjoint domain of the diagonal N precisely when its coefficientwise product by the real sequence $|p|$ is square summable; this proves self-adjointness on the given domain. The truncations by length prove finite supports are a core. Resolvent coefficients $(|p| - i)^{-1}$ tend to zero and each length space is finite dimensional, giving compactness. Prefix and deletion shift the length by at most one, so preserve the domain and prove the commutator formulas there. Leibniz then handles the dense polynomial *-algebra.

Every terminal sector has one vacuum and two paths at each length $n \geq 1$, even when some graph endpoints coincide: distinct formal paths remain distinct basis vectors. Thus restriction to a sector still has a simple vacuum, but it need not be a faithful representation of the whole concrete algebra. The trace of e^{-tN^2} is $E + 2E\sum_{n\geq 1} e^{-tn^2}$, distinct from the semigroup trace of N in the core. The trace identity with $\chi(N)$ is only on $\operatorname{Re} s > 1$, where the product is trace class. Analytic continuation of the scalar L-function does not make that same operator product trace class on $\operatorname{Re} s > 0$.

E Abel determinant estimates and the arithmetic program

For completeness the inserted family has a full elementary convergence proof. Since complete nonprincipal character periods sum to zero, $|A_n| \leq B_0 \leq q_\chi$. On $2^k \leq n < 2^{k+1}$,

$$\sum |u_n|^2 \leq \frac{B_0^2}{(k+1)^2}, \qquad \sum |v_n(s)|^2 \leq M^2(k+1)^2 r^k, \quad r = 2^{-2\sigma_0} < 1.$$

The second follows from $n^{-s} - (n+1)^{-s} = s\int_n^{n+1} x^{-s-1}\,dx$. Define

$$Q_J(r) = r^J\left(\frac{(J+1)^2}{1-r} + \frac{2(J+1)r}{(1-r)^2} + \frac{r(1+r)}{(1-r)^3}\right).$$

Expanding $(J+j+1)^2$ in the three geometric moment sums proves $Q_J = \sum_{k\geq J}(k+1)^2 r^k \to 0$. Locally uniform ℓ^2 convergence of the holomorphic finite-coordinate vectors proves Banach holomorphy. The linear functional $z \mapsto \sum v_n z_n$ has norm $\|v\|$; this is the complex-linear coefficient pairing, not a conjugate-linear dependence on s.

Write $\mathcal{I}(v)(z) = \sum_n v_n z_n$ for the coefficient functional, distinct from the arrow ideal J. For $N_J = 2^J - 1$, $J \geq 1$, let $K_N = u \otimes (\mathcal{I}(P_N v) - e_1^*)$. Then

$$\|K - K_{N_J}\|_1 \leq \sqrt{2} B_0 M \sqrt{Q_J(r)}.$$

Finite summation by parts gives

$$T_N = \det_F(I + K_N) = \sum_{n\leq N} A_n(n^{-s} - (n+1)^{-s}) = S_N - A_N(N+1)^{-s}.$$

Adding the last endpoint times e_1^* to the functional recovers S_N exactly because $u_1 = 1$. Its trace-norm error is at most

$$\sqrt{2} B_0(M\sqrt{Q_J(r)} + B_0(N_J+1)^{-\sigma_0}).$$

For arbitrary cutoffs use the last whole dyadic block for an enclosing tail. Galerkin compressions also converge: $\|u - P_{N_J}u\| \leq B_0/\sqrt{J}$. With $C_K = 1 + M\sqrt{(1+r)/(1-r)^3}$,

$$\|K - P_{N_J} K P_{N_J}\|_1 \leq \frac{B_0 C_K}{\sqrt{J}} + \sqrt{2} B_0 M \sqrt{Q_J(r)}.$$

The raw endpoint adds the same endpoint bound as before. Isometric insertion by U preserves nonzero singular values, every norm estimate, and the Fredholm determinant. Everything is in one compact ideal of the actual infinite dessin Fock algebra.

For primitive parity a use $g(s) = (q_\chi/\pi)^{(s+a)/2}\Gamma((s+a)/2)$, holomorphic nonzero on $\operatorname{Re} s > 0$. The completed functional is $g(s)\mathcal{I}(v) - e_1^*$, and the completed raw functional is $g(s)(\mathcal{I}(P_N v) + A_N(N+$

$1)^{-s}e_1^*) - e_1^*$. Their determinants are exactly gL and gS_N. Scaling the entire old perturbation would instead give $1 + g(L-1)$, the wrong normalization. Trace-class background is standard [26], while these special rank-one identities follow directly from $\det_F(I + u \otimes \varphi) = 1 + \varphi(u)$.

The metric obstruction is likewise elementary. With inner product conjugate-linear in its first variable, symmetry of $T = u \otimes \varphi$ implies

$$\varphi(z)\langle u, u\rangle = \overline{\varphi(u)}\langle u, z\rangle.$$

A nonzero φ cannot have $\varphi(u) = 0$; thus all symmetric maps with that range have proportional functionals. For $n \geq 2$ the coefficient ratios at $\sigma + it$ and σ satisfy

$$\frac{v_n(\sigma + it)}{v_n(\sigma)} = \frac{\sigma + it}{\sigma}\, n^{-it}(1 + O_{\sigma,t}(1/n)).$$

Integers nearest $\exp(2\pi k/|t|)$ and $\exp((2\pi k + \pi)/|t|)$ give opposite nonzero subsequential limits. Hence no common positive metric symmetrizes this family for all critical heights. Restricting any proposed metric on the inserted family to $U\ell^2$ gives the same contradiction. An independently constructed positive comparison is still a possible different route.

E.1 Precisely how classical and recent sources enter

Over a finite field the Grothendieck trace-formula architecture writes a zeta function as alternating determinants on finite-dimensional cohomology; Deligne's purity supplies the weights of eigenvalues [22, 20, 21]. If a weight-w Frobenius is $q^{w/2}$ times a positive-metric unitary, each eigenvalue has modulus $q^{w/2}$. The norm conclusion is immediate; constructing that form or proving purity is the deep step. Symplectic/alternating duality $F^t J F = q^w J$ alone only pairs reciprocal eigenvalues and does not give their individual moduli. An order resolution or Schur-positive character expansion alone cannot replace this step in number-field cohomology.

The recent primary comparisons used here are specific:

- [53, Theorem3.12, pp.15–17] gives finite abelian covers with local orbit length $\log p$ and genuine arithmetic monodromy. It suggests what a descent-equipped order functor should match; it does not identify our u or a face cycle with a prime without such a functor.

- [54, Propositions4.1–4.3, Theorem4.6] builds finite-prime local models with archimedean Fourier factors. Its Hilbertian isomorphisms are not an asserted uniformly controlled common isometry for our arithmetic approximants. [55, Proposition3.1] gives determinacy of a particular local moment problem; positive finite Hankel matrices are not the missing global Weil inequality.

- [56, Theorem3.6] gives discreteness/lower boundedness for the stated restricted Weil form. Theorem5.10, pp.23–24, additionally assumes a finite truncation, $\lambda > 1$, a simple even bottom eigenfunction ξ and $\delta_N \xi = 1$. Its quotient metric is positive and its regularized determinant is $-i\lambda^{-iz}\widehat{\xi}(z)$. Passing to the infinite arithmetic comparison, including the ground-state and convergence conditions in its §8, is not automatic.

- [57, Theorem6.1, p.13] assumes a specified real distribution on an interval, the corresponding trigonometric-core form, essential self-adjointness, and an isolated simple lowest eigenvalue with reflection-even eigenfunction. Its Fourier transform then has only real zeros. Our degree operator's vacuum and length parity are not those data.

- [60, Theorem6.1] gives prime-indexed periodic packets in its dynamical construction, not one proved orbit per prime with the full desired cohomological trace formula. [62] isolates an obstruction to a particular real-coefficient cohomology with full functorial Artin multiplicities; it is not a theorem excluding every Hilbert–Pólya operator. The current survey [58] distinguishes remaining steps. The recent Picard-monoid construction [59] is not already a polarized abelian variety with a number-field purity theorem. [63] uses RH as an input in the cited spectral identification, so cannot be used circularly as its proof.

These are targeted theorem/proof readings, not claims that every page of every cited work has been audited. The preserved review records specify actual page coverage and hashes, including inaccessible original PDFs where the modern primary proof was used instead.

E.2 A concrete sequence of remaining constructions

First choose a number-field model of a Belyi pair, its finite covers and conductor/ramification data. Construct actions or exact correspondences on the labelled order/module or analytic categories that recover the local prime Frobenius polynomials. The elliptic twists show why omitting this datum loses information. Test at more than one arithmetic model of the same complex dessin, not just its topological passport.

Second specify a finite projective or otherwise determinant-admissible complex and its *independent* arithmetic endomorphism. Prove exact-sequence/determinant-line compatibility. Use semi-invariants to encode presentation degeneracies, and Schur/exterior characters to calculate local factors, retaining every central-action and nilpotence distinction above. General derived Schur replacements require their own construction; termwise Schur on the resolution fails the exhibited split sequence test.

Third prove the same local factors, gamma terms and conductor give a globally convergent or correctly regularized determinant. Trace-norm convergence must be established for the chosen analytic perturbations; strong convergence is insufficient in general. The inserted Abel family supplies this topology for an elementary scalar encoding, but has neither the independent arithmetic spectrum nor the positive metric needed for zero location.

Finally produce an independently proved positive form or self-adjoint determinant-preserving realization. One acceptance criterion is the Hurwitz target in the core; another is equality with the precise arithmetic quadratic form in the recent spectral theorems, with all their domain, ground-state, normalization and limit hypotheses verified. The local Cauchy kernel, K-group computation and graph spectral dimension alone meet none of these final comparison hypotheses. No remaining condition is silently asserted true here. Dirichlet, Hecke, Dedekind and Artin scopes must remain individually recorded.

F Literal formal and computational coverage

`DessinNodeOrder.lean` defines the fiber-product subring of two genuine power-series rings. Its twenty printed declarations establish diagonal injectivity, nonzero branches, branch zero products, factorizations and annihilator identities, regular u, augmentation surjectivity and exactness, bijective power-series coordinates, square-zero alternating maps, three kernel-equals-range statements and projectivity of the actual regular/product terms. The coordinate statement is a bijection; the file does not yet package it as an R-linear equivalence or declare the general order freeness theorem.

`DessinInvariantComponents.lean` has ten printed declarations: three CK1-collapse components, determinant change of bases with an actual inverse, symmetric-square zero, identity,

composition and relation preservation, the degree-two Schur character identity, and the nodal central-parameter obstruction. Its symmetric-square matrix is

$$\mathrm{Sym}^2\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 & ab & b^2 \\ 2ac & ad+bc & 2bd \\ c^2 & cd & d^2 \end{pmatrix}.$$

This is the ordinary monomial basis convention; composition is proved universally over any commutative ring. The full project suite prints types and foundational dependencies; no theorem about general LR decompositions, abstract derived equivalences, C* K-theory or GRH is inferred from these component proofs.

The exact Python checks use fixed rational matrices and exhaustive finite sets, with no random seeds. They construct the displayed permutations and paths, evaluate rational Cartan identities, check formal Euler products through a stated degree, verify unrestricted-output Fock basis identities on bounded input sets, compute incidence Smith data, construct all nine string representations and signed maps, and test Young/LR/Cauchy examples and the seven-element-field counts. Those diagnostics are reproducible finite corroboration. The universal infinite claims rest on the proofs or explicitly cited theorems, and the arithmetic comparison remains open.

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